

Pharma Validation Library — Phase 1

AgenThink Mesh Retrospective Validation Corpus

Document Classification: Board-Level Research Report **Audience:** FDA/EMA Reviewers, Pharmaceutical R&D Leadership, CRO Executives, Big Four Diligence Teams, Sovereign Wealth Fund Investment Committees **Prepared by:** AgenThink Mesh Governance Research Unit **Date:** June 2026 **Version:** 1.0

Prefatory Note

This document does not make promotional claims. It presents a methodology, a ranked corpus, and a statistical framework for evaluating whether a governed multi-persona council — operating exclusively on pre-decision evidence — consistently identifies the key risk signals, opportunity signals, and decision drivers that were visible before major pharmaceutical outcomes became known.

The standard applied throughout is scientific defensibility, not commercial utility. Every claim in this document is grounded in published literature, regulatory filings, or documented financial disclosures. Where evidence is incomplete, that incompleteness is stated explicitly.

The audience for this work is assumed to be skeptical. That skepticism is appropriate. The value of retrospective validation is proportional to the rigor of its methodology, and rigor requires honesty about what the data can and cannot prove.

Executive Summary

Phase II to Phase III advancement decisions are among the most consequential and least governed decisions in pharmaceutical R&D. The average Phase III failure rate for drugs that reach Phase III is approximately 40–50%, with cardiovascular and neurological indications performing substantially worse. The estimated cost of a failed Phase III program ranges from 300M to 2B in direct R&D expenditure, with market capitalization impacts that can reach \$20B or more for large-cap companies.

This report identifies 20 historical Phase II to Phase III advancement decisions suitable for retrospective council analysis, ranks them by suitability, recommends the 10 highest-priority cases for immediate execution, and defines the statistical framework required before any public claims about council accuracy can be made.

The central question is not whether a governed council would have made the right call in every case. The central question is whether a governed council, operating on pre-decision evidence, would have identified the key signals that determined the outcome. That is a testable, falsifiable claim — and this report defines the methodology for testing it.

Deliverable A — Top 20 Ranked Cases

Methodology

Each case was scored on six dimensions, each rated 1–10:

Dimension	Definition
Evidence Quality (EQ)	Completeness and public availability of pre-decision Phase II evidence
Decision Complexity (DC)	Difficulty of the Phase II to III advancement decision
Financial Significance (FS)	Magnitude of financial impact (positive or negative)
Governance Relevance (GR)	Relevance to decision governance failures or successes
Pre-Decision Availability (PA)	Availability of evidence that was public before Phase III
Educational Value (EV)	What this case teaches about Phase II to III decision quality

The composite score is the unweighted average of all six dimensions. Cases are ranked by composite score descending.

Top 20 Ranked Table

Rank	Drug	Company	Therapeutic Area	Ph2	Adv	EQ	DC	FS	GR	PA	EV	Score	Type
1	BNT162b2 (Comirnaty)	Pfizer/BioNTech	Infectious Disease — COVID-19 mRNA vaccine	2020	2020	9	8	10	9	8	10	9.2	Bloc Laur
2	Pembrolizumab (Keytruda)	Merck	Oncology — PD-1 inhibitor, melanoma	2014	2014	9	8	10	9	9	10	9.2	Bloc Laur
3	Sofosbuvir (Sovaldi)	Gilead Sciences	Hepatology — HCV NS5B inhibitor	2011	2012	9	8	10	9	8	10	9.0	Bloc Laur
4	Ivacaftor (Kalydeco)	Vertex Pharmaceuticals	Respiratory — CFTR potentiator, CF G551D	2010	2009	10	8	10	9	8	10	9.0	Bloc Laur
5	Crizotinib (Xalkori)	Pfizer	Oncology — ALK/ROS1 inhibitor, NSCLC	2010	2011	9	7	8	9	9	9	8.5	Bloc Laur
6	Semagacestat (LY450139)	Eli Lilly	Neurology — gamma-secretase inhibitor, AD	2008	2009	8	8	9	9	8	9	8.5	Term Prog
7	Nusinersen (Spinraza)	Ionis/Biogen	Neurology — antisense, SMA	2015	2016	9	7	9	8	8	9	8.3	Bloc Laur
8	Solanezumab	Eli Lilly	Neurology — anti-amyloid antibody, AD	2012	2013	7	8	9	9	7	9	8.2	Majc
9	Torcetrapib (CP-529,414)	Pfizer	Cardiovascular — CETP inhibitor	2004	2005	8	8	9	8	6	9	8.0	Majc
10	Ticagrelor (Brilinta)	AstraZeneca	Cardiovascular — P2Y12 inhibitor, ACS	2007	2006	8	8	9	8	6	9	8.0	Bloc Laur
11	Entrectinib (Rozlytrek)	Ignyta/Roche	Oncology — ROS1/TRK inhibitor, NTRK fusions	2017	2017	8	7	9	8	7	9	8.0	Bloc Laur
12	Evolocumab (Repatha)	Amgen	Cardiovascular — PCSK9 inhibitor	2012	2013	8	8	9	7	7	9	8.0	Bloc Laur
13	Muraglitazar	BMS/AstraZeneca	Metabolic — dual PPAR agonist, T2D	2004	2005	8	8	8	9	8	9	8.0	Term Prog
14	Evacetrapib	Eli Lilly	Cardiovascular — CETP	2013	2013	8	7	7	9	7	9	7.8	Majc

Rank	Drug	Company	Therapeutic Area	Ph2	Adv	EQ	DC	FS	GR	PA	EV	Score	Type
			inhibitor										
15	Eteplirsen (Exondys 51)	Sarepta Therapeutics	Neurology — exon 51 skipping, DMD	2013	2016	5	9	8	9	4	9	7.3	Cont Appr
16	Idelalisib (Zydelig)	Gilead Sciences	Oncology — PI3K-delta inhibitor, CLL/NHL	2013	2013	8	8	7	9	6	9	7.3	Cont Appr
17	Rosiglitazone (Avandia)	GlaxoSmithKline	Metabolic — PPAR-gamma agonist, T2D	1997	1999	6	7	9	9	4	9	7.3	Regu Rev
18	Aducanumab (Aduhelm)	Biogen/Eisai	Neurology — anti-amyloid antibody, AD	2016	2014	6	9	9	9	6	9	7.2	Cont Appr
19	Dalcetrapib	Roche	Cardiovascular — CETP inhibitor	2009	2011	8	8	8	7	6	9	7.2	Term Prog
20	Verubecestat (MK-8931)	Merck	Neurology — BACE1 inhibitor, AD	2015	2015	8	8	7	7	6	9	7.0	Term Prog

Case Type Distribution

Type	Count	Percentage
Blockbuster Launch	8	40%
Major Failure	3	15%
Terminated Program	4	20%
Controversial Approval	3	15%
Regulatory Reversal	1	5%
Major Success	1	5%

The corpus covers the full spectrum of Phase III outcomes, providing a balanced test of council accuracy across both positive and negative decision scenarios.

Deliverable B — Top 10 Cases for Immediate Execution

Selection Criteria

The Top 10 selection must satisfy four mandatory composition requirements:

- At least 3 major successes
- At least 3 major failures
- At least 2 controversial cases
- At least 2 regulatory edge cases

The selection also prioritises cases where pre-decision evidence is most complete and publicly available, as these cases provide the cleanest test of council signal detection.

Top 10 Recommended Execution List

Priority	Drug	Type	Rationale
1	Torcetrapib	Major Failure	Pilot already completed. Retrospective validation confirmed. Serves as anchor case.
2	Semagacestat	Major Failure	Gamma-secretase inhibitor with worsening cognition signal in Phase II. Clean pre-decision evidence. Tests amyloid hypothesis governance.
3	Muraglitazar	Terminated Program / Regulatory Edge	CV safety signal in Phase II dismissed. FDA advisory committee flagged it. Withdrawn pre-approval. Tests safety signal governance.
4	Sofosbuvir	Major Success	Transformative HCV cure. Clean Phase II data. Tests council ability to identify breakthrough signals and avoid false-positive blocking.
5	Pembrolizumab	Major Success / Blockbuster	PD-1 inhibitor with strong Phase II signal. Tests council recognition of paradigm-shifting mechanisms.
6	Aducanumab	Controversial Approval	Mixed Phase III results. FDA approval over advisory committee objection. Tests council handling of surrogate endpoints and regulatory controversy.
7	Eteplirsen	Controversial Approval / Regulatory Edge	12-patient Phase II study. Accelerated approval over strong internal FDA opposition. Tests evidence threshold governance.
8	Rosiglitazone	Regulatory Reversal	Approved 1999, restricted 2010. CV safety signal visible in post-marketing data. Tests long-horizon governance.
9	Ivacaftor	Major Success	Precision medicine in rare disease. Exceptional Phase II signal. Tests council recognition of high-confidence precision signals.
10	Evacetrapib	Major Failure	Third CETP inhibitor failure. Tests whether council learns from class precedent (torcetrapib, dalcetrapib) to identify futility risk.

Composition Verification

Requirement	Cases	Met
At least 3 major successes	Sofosbuvir (#4), Pembrolizumab (#5), Ivacaftor (#9)	Yes
At least 3 major failures	Torcetrapib (#1), Semagacestat (#2), Evacetrapib (#10)	Yes
At least 2 controversial cases	Aducanumab (#6), Eteplirsen (#7)	Yes
At least 2 regulatory edge cases	Muraglitazar (#3), Eteplirsen (#7), Rosiglitazone (#8)	Yes

All four composition requirements are satisfied.

Deliverable C — Pharma Validation Scorecard

Purpose

The Pharma Validation Scorecard provides a standardised framework for evaluating the accuracy of each retrospective council run. It is designed to produce a single, auditable score that can be compared across cases and used to assess the statistical validity of the council's signal detection capability.

The scorecard is not a marketing instrument. It is a scientific measurement tool. Every dimension is defined operationally, with explicit criteria for each score level.

Scorecard Dimensions

Dimension 1: Council Verdict Accuracy (CVA)

Definition: Did the council's verdict (GO / WAIT / NO-GO) match the retrospectively correct decision?

Score	Criteria
10	Council verdict exactly matches retrospectively correct decision
7	Council verdict partially correct (e.g., WAIT when NO-GO was correct, but key blocker identified)
4	Council verdict incorrect but key signal identified in rationale
0	Council verdict incorrect and key signal not identified

Dimension 2: Actual Outcome Alignment (AOA)

Definition: How well did the council's predicted outcome align with the actual Phase III outcome?

Score	Criteria
10	Council predicted the specific failure mode or success mechanism
7	Council predicted the direction (failure/success) but not the specific mechanism
4	Council predicted mixed outcome
0	Council predicted opposite direction

Dimension 3: Signal Detection Accuracy (SDA)

Definition: What fraction of the key pre-decision signals did the council identify?

Calculated as: $(\text{Signals correctly identified} / \text{Total key signals present}) \times 10$

Key signals are defined as those documented in the primary Phase II publication or regulatory filing.

Dimension 4: Correct Blocker Identification (CBI)

Definition: For failure cases — did the council identify the specific mechanism that caused Phase III failure? For success cases — did the council identify the specific mechanism that drove Phase III success?

Score	Criteria
10	Primary blocker/driver correctly identified with mechanistic specificity
7	Primary blocker/driver identified at category level (e.g., "safety signal" without specific mechanism)
4	Secondary blocker/driver identified; primary missed
0	No relevant blocker/driver identified

Dimension 5: False Positive Blockers (FPB)

Definition: For success cases — did the council raise blockers that would have incorrectly prevented a successful drug from advancing?

Score	Criteria
10	No false positive blockers raised
7	Minor false positive blockers raised but not decision-determinative
4	Significant false positive blockers raised that would have delayed advancement
0	False positive blockers would have prevented advancement of a successful drug

Dimension 6: Missed Risks (MR)

Definition: What fraction of the key risk signals present in Phase II data did the council fail to identify?

Calculated as: $(1 - (\text{Missed risks} / \text{Total risks present})) \times 10$

Dimension 7: Governance Quality (GQ)

Definition: Did the council apply the correct constitutional rules? Were the right personas most influential? Was the deliberation process sound?

Score	Criteria
10	All relevant constitutional rules applied; correct personas drove the decision; deliberation was sound
7	Most relevant rules applied; minor gaps in persona influence
4	Some rules applied; significant gaps
0	Constitutional rules not applied; deliberation unsound

Dimension 8: Evidence Completeness (EC)

Definition: Did the council use all available pre-decision evidence? Were there material evidence gaps?

Score	Criteria
10	All material pre-decision evidence used; no gaps
7	Most material evidence used; minor gaps
4	Significant evidence gaps
0	Material evidence not used

Dimension 9: Retrospective Validation Score (RVS)

Definition: Composite score — weighted average of all eight dimensions.

Weighting:

Dimension	Weight
Council Verdict Accuracy	20%
Signal Detection Accuracy	20%
Correct Blocker Identification	20%
False Positive Blockers	15%
Missed Risks	10%
Governance Quality	5%
Evidence Completeness	5%
Actual Outcome Alignment	5%

$RVS = (CVA \times 0.20) + (SDA \times 0.20) + (CBI \times 0.20) + (FPB \times 0.15) + (MR \times 0.10) + (GQ \times 0.05) + (EC \times 0.05) + (AOA \times 0.05)$

Scorecard Template

Dimension	Weight	Score (0-10)	Weighted Score
Council Verdict Accuracy (CVA)	20%	—	—
Signal Detection Accuracy (SDA)	20%	—	—
Correct Blocker Identification (CBI)	20%	—	—
False Positive Blockers (FPB)	15%	—	—
Missed Risks (MR)	10%	—	—
Governance Quality (GQ)	5%	—	—
Evidence Completeness (EC)	5%	—	—
Actual Outcome Alignment (AOA)	5%	—	—
Retrospective Validation Score (RVS)	100%	—	—/10

Torcetrapib Pilot Scorecard (Completed)

Dimension	Weight	Score	Weighted
Council Verdict Accuracy	20%	10	2.00
Signal Detection Accuracy	20%	10	2.00
Correct Blocker Identification	20%	10	2.00
False Positive Blockers	15%	10	1.50
Missed Risks	10%	9	0.90
Governance Quality	5%	9	0.45
Evidence Completeness	5%	9	0.45
Actual Outcome Alignment	5%	10	0.50
RVS	100%	—	9.8 ₁₀

Notes: The council correctly identified the BP signal as the primary blocker, correctly predicted the failure mechanism (off-target cardiovascular effect), and issued a WAIT verdict that was retrospectively correct. The 0.1 deduction in Missed Risks reflects the fact that the council did not specifically predict the aldosterone mechanism — it identified the BP signal as unexplained and molecule-specific, but did not name aldosterone as the likely pathway (which was not published until 2008). This is appropriate: the council was operating on pre-2006 evidence, and the aldosterone mechanism was not documented until Forrest et al. 2008.

Deliverable D — Validation Roadmap

Execution Sequence

The 10 cases are sequenced to maximise learning velocity and statistical coverage. The sequence moves from the highest-confidence cases (where council accuracy is most likely) to the most challenging cases (where council accuracy is most uncertain), building a credible track record before testing edge cases.

Case 1: Torcetrapib — COMPLETED

Why selected: Highest composite score among failure cases. Clean pre-decision evidence. BP signal is mechanistically specific and epidemiologically quantifiable. Serves as the anchor case for all subsequent validation.

What we expect to learn: Whether the council correctly identifies a specific, quantifiable safety signal as a Phase III blocker when financial pressure to advance is documented and the company characterised the signal as “small.”

Personas most tested: Drug Safety Expert, Chief Biostatistician, Clinical Pharmacologist.

Constitutional rules most challenged: PC-002 (Safety Signal Priority), PC-003 (Financial Independence), PC-007 (Mechanistic Investigation Requirement).

Result: WAIT — 0 GO / 8 WAIT / 2 NO-GO. RVS: $9.\frac{8}{10}$. Retrospectively correct.

Case 2: Semagacestat (Eli Lilly, 2009)

Why selected: Gamma-secretase inhibitor with worsening cognition signal in Phase II. Lilly advanced to Phase III despite the signal. Phase III terminated in 2011 for worsening cognition and increased skin cancer risk. Clean pre-decision evidence. Tests whether the council identifies a cognitive worsening signal in a neurological indication — the most sensitive possible safety signal in the target population.

What we expect to learn: Whether the council correctly identifies a cognitive worsening signal as a hard blocker in an Alzheimer's disease indication, where the patient population is the most vulnerable to cognitive harm.

Personas most tested: Drug Safety Expert, Patient Advocate, Clinical Pharmacologist.

Constitutional rules most challenged: PC-002 (Safety Signal Priority), PC-007 (Mechanistic Investigation), PC-010 (Dissent Documentation).

Case 3: Muraglitazar (BMS/AstraZeneca, 2005)

Why selected: Dual PPAR agonist with cardiovascular safety signal in Phase II. FDA advisory committee flagged the signal. Drug was withdrawn pre-approval. Tests whether the council identifies a cardiovascular safety signal in a metabolic indication — a pattern that recurs across multiple cases (torcetrapib, rosiglitazone, muraglitazar).

What we expect to learn: Whether the council identifies a cardiovascular safety signal in a non-cardiovascular indication (type 2 diabetes), where the signal may be characterised as an incidental finding rather than a primary concern.

Personas most tested: Regulatory Strategist, Drug Safety Expert, Devil's Advocate.

Constitutional rules most challenged: PC-002 (Safety Signal Priority), PC-005 (Competitive Class Analysis), PC-009 (Regulatory Risk Documentation).

Case 4: Sofosbuvir (Gilead, 2012)

Why selected: Transformative HCV cure with exceptional Phase II data. Tests the council's ability to identify a breakthrough signal and avoid false-positive blocking. The risk in this case is not a safety signal — it is the risk of over-caution in a high-unmet-need indication.

What we expect to learn: Whether the council correctly issues a GO verdict for a drug with clean Phase II data, strong mechanistic rationale, and high unmet need — without raising false-positive blockers that would have delayed a transformative therapy.

Personas most tested: Commercial Assessor, Portfolio Manager, Patient Advocate.

Constitutional rules most challenged: PC-001 (Evidence Primacy), PC-004 (Surrogate Endpoint Scrutiny — in this case, SVR12 as a surrogate for cure).

Case 5: Pembrolizumab (Merck, 2014)

Why selected: First PD-1 inhibitor to receive accelerated approval. Paradigm-shifting mechanism. Tests whether the council recognises a novel mechanism of action as a breakthrough signal, even when the Phase II dataset is small (KEYNOTE-001 had 135 patients).

What we expect to learn: Whether the council correctly identifies a paradigm-shifting mechanism and issues a GO verdict despite a small Phase II dataset, correctly applying the accelerated approval pathway logic.

Personas most tested: Scientific Skeptic, Regulatory Strategist, Commercial Assessor.

Constitutional rules most challenged: PC-001 (Evidence Primacy — small dataset), PC-004 (Surrogate Endpoint — ORR as surrogate for OS), PC-009 (Regulatory Risk Documentation).

Case 6: Aducanumab (Biogen/Eisai, 2019)

Why selected: Most controversial FDA approval in recent memory. Phase III results were mixed — one trial positive, one negative. FDA approved over advisory committee objection (10-0-1 against). Tests the council's handling of contradictory Phase III data and

surrogate endpoint controversy.

What we expect to learn: Whether the council correctly identifies the contradictory Phase III data as a WAIT or NO-GO signal, and whether it correctly applies the surrogate endpoint rule (amyloid plaque reduction as a surrogate for cognitive benefit).

Personas most tested: Regulatory Strategist, Scientific Skeptic, Devil's Advocate.

Constitutional rules most challenged: PC-004 (Surrogate Endpoint Scrutiny), PC-006 (Quantitative Risk-Benefit), PC-009 (Regulatory Risk Documentation), PC-010 (Dissent Documentation).

Case 7: Eteplirsen (Sarepta, 2016)

Why selected: Accelerated approval based on a 12-patient Phase II study. Strong FDA internal opposition. Tests the council's handling of an extremely small evidence base in a rare disease with high unmet need and strong patient advocacy pressure.

What we expect to learn: Whether the council correctly identifies the evidence threshold issue (12 patients) as a WAIT signal requiring a larger confirmatory study, or whether it issues a GO verdict based on unmet need and patient advocacy pressure.

Personas most tested: Chief Biostatistician, Patient Advocate, Regulatory Strategist.

Constitutional rules most challenged: PC-001 (Evidence Primacy), PC-003 (Financial Independence), PC-008 (DSMB Charter Alignment).

Case 8: Rosiglitazone (GSK, 1999)

Why selected: Approved 1999, restricted 2010. Cardiovascular safety signal emerged post-marketing. Tests the council's ability to identify long-horizon risk signals that were not visible in Phase II but were visible in the regulatory filing and early post-marketing data.

What we expect to learn: Whether the council correctly identifies the cardiovascular risk signal in a metabolic indication, and whether it correctly applies the long-horizon risk framework for a drug targeting a chronic disease population.

Personas most tested: Drug Safety Expert, Regulatory Strategist, Portfolio Manager.

Constitutional rules most challenged: PC-002 (Safety Signal Priority), PC-006 (Quantitative Risk-Benefit), PC-009 (Regulatory Risk Documentation).

Case 9: Ivacaftor (Vertex, 2012)

Why selected: Exceptional Phase II signal in a precision medicine indication. Tests the council's ability to recognise a high-confidence precision signal and issue a clean GO verdict without false-positive blocking.

What we expect to learn: Whether the council correctly issues a GO verdict for a drug with a validated biomarker (CFTR G551D mutation), exceptional Phase II efficacy (FEV1 improvement of 10.6 percentage points vs placebo), and a well-characterised safety profile.

Personas most tested: Commercial Assessor, Portfolio Manager, Patient Advocate.

Constitutional rules most challenged: PC-001 (Evidence Primacy — small rare disease dataset), PC-004 (Surrogate Endpoint — FEV1 as surrogate for clinical benefit).

Case 10: Evacetrapib (Eli Lilly, 2015)

Why selected: Third CETP inhibitor failure after torcetrapib (2006) and dalcetrapib (2012). Tests whether the council learns from class precedent and correctly identifies futility risk in a drug with a clean safety profile but a failed class history.

What we expect to learn: Whether the council correctly identifies the class precedent (two prior CETP inhibitor failures) as a primary blocker, even when the drug's own Phase II data appears clean. This tests the council's application of PC-005 (Competitive Class Analysis).

Personas most tested: Chief Biostatistician, Scientific Skeptic, Portfolio Manager.

Constitutional rules most challenged: PC-004 (Surrogate Endpoint — HDL-C as surrogate for CV benefit), PC-005 (Competitive Class Analysis), PC-006 (Quantitative Risk-Benefit).

Deliverable E — Statistical Credibility Assessment

Prefatory Caution

This section is written for an audience that includes FDA reviewers, EMA reviewers, and academic statisticians. The standards applied are those of peer-reviewed clinical research, not those of commercial validation. Claims are graded by the level of evidence they

require, and the minimum case count for each claim level is stated explicitly.

The fundamental challenge is that retrospective validation is not a randomised controlled trial. The council cannot be blinded to the drug name, company, and therapeutic area — all of which carry information about the likely outcome. This is a structural limitation of retrospective validation that cannot be fully resolved. It can be mitigated through methodology (evidence cutoff enforcement, persona independence, constitutional rule pre-commitment), but it cannot be eliminated.

Any claims made on the basis of retrospective validation must acknowledge this limitation explicitly.

Statistical Framework

Base Rate Context

The relevant base rate for council accuracy is not 50% (random guessing). The relevant base rate is the accuracy of the actual Phase II to III advancement decisions made by the companies in this corpus. Of the 20 cases in this library:

- 8 were blockbuster launches (companies were correct to advance)
- 4 were major failures or terminated programs where the company advanced despite visible warning signals (companies were incorrect to advance)
- 3 were controversial approvals (outcome is contested)
- 3 were regulatory reversals or edge cases (outcome is mixed)
- 2 were major failures with no visible pre-decision warning signals (companies were correct to advance given available evidence)

The base rate for “company made the correct advancement decision given available pre-decision evidence” is approximately 50–60% across this corpus. A council that simply replicated the company’s decision would score 50–60% accuracy. The council must demonstrably outperform this base rate to make a credible claim.

Credibility Thresholds by Case Count

After 1 case (current status: Torcetrapib completed)

Defensible claims:

- “We have run one retrospective pilot. The council issued a WAIT verdict on torcetrapib. The actual outcome was a Phase III termination for excess mortality. The council’s primary blocker (unresolved BP signal) was confirmed as the cause of failure by Forrest et al. 2008.”
- “This is a proof of concept, not a validation. One case proves that the methodology can be applied. It does not prove that the methodology is accurate.”

Non-defensible claims:

- Any claim about accuracy rates, sensitivity, or specificity
- Any claim about the council’s ability to generalise across therapeutic areas or decision types

After 5 cases

Defensible claims:

- Descriptive statistics: “The council correctly identified the primary blocker in X of 5 cases.”
- Directional accuracy: “The council issued the retrospectively correct verdict (GO/WAIT/NO-GO) in X of 5 cases.”
- Qualitative pattern: “Across 5 cases, the council consistently [identified safety signals / avoided false-positive blocking / applied constitutional rules].”

Non-defensible claims:

- Statistical significance (5 cases is insufficient for any inferential statistics)
- Generalisation to untested therapeutic areas
- Comparison to industry base rates (requires matched controls)

After 10 cases

Defensible claims:

- Point estimates with confidence intervals: “The council correctly identified the primary blocker in 8 of 10 cases (80%, 95% CI: 44–97%).” Note: confidence intervals at n=10 are very wide.

- Subgroup patterns: “In failure cases (n=4), the council correctly identified the primary blocker in all 4 cases. In success cases (n=4), the council issued a GO verdict without false-positive blocking in 3 of 4 cases.”
- Comparison to a stated base rate: “The company’s actual advancement decision was retrospectively correct in 6 of 10 cases (60%). The council’s verdict was retrospectively correct in 8 of 10 cases (80%).”

Non-defensible claims:

- Statistical significance ($p < 0.05$ requires approximately 20+ cases for a meaningful test)
- Claims about specific therapeutic areas (subgroup sizes are too small)
- Claims about prospective accuracy (retrospective validation does not prove prospective accuracy)

After 20 cases

Defensible claims:

- Statistical significance: With 20 cases, a binomial test can detect accuracy rates significantly above 50% with reasonable power. If the council achieves 80% accuracy ($\frac{16}{20}$), the p-value against a 50% null hypothesis is approximately 0.006.
- Subgroup analysis: “In cardiovascular failure cases (n=5), the council correctly identified the primary blocker in X of 5 cases.”
- Comparison to base rate: “The council’s retrospective accuracy (X/20) is significantly above the base rate of company advancement decision accuracy (approximately 60%).”
- Peer-reviewed publication: 20 cases is the minimum for a credible methods paper in a peer-reviewed journal. The paper would describe the methodology, present the 20 cases, and report accuracy metrics with confidence intervals.

Non-defensible claims:

- Claims about prospective accuracy
- Claims about specific rare disease indications (insufficient case count)
- Claims about superiority to any specific existing process (requires head-to-head comparison)

After 50 cases

Defensible claims:

- Robust statistical analysis: 50 cases provides sufficient power for subgroup analysis by therapeutic area, decision type, and outcome category.
- Sensitivity and specificity: “The council’s sensitivity for identifying Phase III failures is X% (95% CI: Y–Z%). The council’s specificity for correctly approving successful drugs is X% (95% CI: Y–Z%).”
- Comparison to industry benchmarks: With 50 cases, a matched comparison to historical industry advancement decision accuracy becomes feasible.
- Peer-reviewed publication in a high-impact journal: 50 cases is sufficient for a substantive methods and results paper in journals such as Nature Medicine, NEJM Evidence, or Clinical Pharmacology and Therapeutics.
- Enterprise pharma buyer consideration: At 50 cases, the evidence base is sufficient for a serious conversation with pharmaceutical R&D leadership about prospective application. It is not sufficient for regulatory submission or clinical trial use.

Non-defensible claims at any case count:

- Claims that the council’s retrospective accuracy proves prospective accuracy
- Claims that the council replaces or supersedes existing regulatory review processes
- Claims about specific drug candidates without running the full council protocol

Summary Table: Credibility by Case Count

Cases	Statistical Claim	Enterprise Buyer Consideration	Peer Review Eligibility	Regulatory Relevance
1	Proof of concept only	Demonstration only	Not eligible	None
5	Descriptive statistics only	Pilot evidence	Not eligible	None
10	Point estimates (wide CI)	Serious pilot evidence	Methods note possible	Background reference only
20	Significant vs. 50% base rate	Credible evidence for early adoption	Full methods paper eligible	Background reference
50	Sensitivity/specificity estimates	Sufficient for enterprise pilot	High-impact journal eligible	Exploratory regulatory discussion

Recommendation

The minimum case count before any public claims about council accuracy should be made is **10 completed cases**, with the following conditions:

1. All 10 cases must use the same constitution version (Pharma-Constitution-v1.0 or later)
2. All 10 cases must have a documented evidence cutoff enforced before council input
3. All 10 cases must have a completed scorecard with all nine dimensions scored
4. The retrospective appendix for each case must be clearly separated from the council input record
5. The methodology must be documented in sufficient detail for independent replication

Before 10 cases are completed, the appropriate claim is: "We have completed X retrospective pilots. The methodology is documented. The results are [described]. We are building toward a statistically meaningful validation corpus."

Appendix: Case Summaries — Ranks 1–20

Rank 1: BNT162b2 (Comirnaty) — Pfizer/BioNTech

Type: Blockbuster Launch | **Score:** $9.\frac{2}{10}$

BNT162b2 was the first mRNA vaccine to receive emergency use authorization for COVID-19. Phase II data (BNT162b2-01, NCT04368728) demonstrated 95% efficacy and an acceptable safety profile in a 44,000-patient Phase III trial. The advancement decision was made under extraordinary time pressure in 2020, with Phase II and Phase III running concurrently under Operation Warp Speed. The governance challenge is the opposite of torcetrapib: the risk was not advancing a drug with a safety signal, but advancing a novel platform technology at unprecedented speed with compressed evidence timelines.

Governance value: Tests the council's ability to issue a GO verdict under time pressure with a novel platform technology, while correctly identifying the specific risks of compressed timelines (long-term safety data, manufacturing scale, cold chain logistics).

Rank 2: Pembrolizumab (Keytruda) — Merck

Type: Blockbuster Launch | **Score:** $9.\frac{2}{10}$

Pembrolizumab received accelerated approval in September 2014 for advanced melanoma based on KEYNOTE-001, a Phase II study of 135 patients with an objective response rate of 26%. It became the first PD-1 inhibitor approved by the FDA. The advancement decision required the council to evaluate a novel mechanism (PD-1 checkpoint inhibition), a small Phase II dataset, a surrogate endpoint (ORR as surrogate for OS), and a high-unmet-need indication. The drug subsequently became the best-selling oncology drug in history, with annual revenues exceeding \$20B.

Governance value: Tests the council's ability to recognise a paradigm-shifting mechanism and issue a GO verdict despite a small Phase II dataset, correctly applying the accelerated approval pathway logic without false-positive blocking.

Rank 3: Sofosbuvir (Sovaldi) — Gilead Sciences

Type: Blockbuster Launch | **Score:** 9.0/10

Sofosbuvir is a nucleotide analogue NS5B inhibitor that achieved sustained virologic response (SVR12) rates of 90%+ in Phase II studies for HCV genotype 1. The advancement decision was straightforward from an efficacy standpoint, but the governance challenge was the pricing and access implications of a 1,000/*pilldrug*. *The Phase II data was exceptionally clean : high SVR12 rates, manageable safety profile, and a validated surrogate endpoint (SVR12 as a surrogate for HCV cure). The drug received* 10.3B in revenue in its first year.

Governance value: Tests the council's ability to issue a clean GO verdict for a drug with exceptional Phase II data, while correctly identifying the access and pricing governance issues that are not typically part of a Phase II to III advancement decision.

Rank 4: Ivacaftor (Kalydeco) — Vertex Pharmaceuticals

Type: Blockbuster Launch | **Score:** 9.0/10

Ivacaftor is a CFTR potentiator for cystic fibrosis patients with the G551D mutation. Phase II data demonstrated a 10.6 percentage point improvement in FEV1 vs placebo — an extraordinary effect size for a chronic respiratory disease. The governance challenge was the small patient population (approximately 4% of CF patients carry the G551D mutation) and the surrogate endpoint (FEV1 as a surrogate for clinical benefit). The drug received FDA approval in January 2012 and demonstrated that precision medicine in rare disease could produce exceptional Phase II signals that translate reliably to Phase III.

Governance value: Tests the council's ability to recognise a high-confidence precision signal in a small patient population and issue a clean GO verdict, correctly applying the rare disease evidence threshold.

Rank 5: Crizotinib (Xalkori) — Pfizer

Type: Blockbuster Launch | **Score:** 8.5/10

Crizotinib is an ALK/ROS1 inhibitor for ALK-positive NSCLC. Phase II data (PROFILE 1001) demonstrated an objective response rate of 57% in ALK-positive NSCLC patients, compared to a historical control ORR of approximately 10% for standard chemotherapy. The advancement decision was made under the FDA's breakthrough therapy designation framework. The drug received accelerated approval in August 2011, less than 5 years after the ALK fusion was first described in NSCLC.

Governance value: Tests the council's ability to recognise a validated biomarker (ALK fusion) as a precision signal and issue a GO verdict with a small Phase II dataset, correctly applying the breakthrough therapy designation logic.

Rank 6: Semagacestat (LY450139) — Eli Lilly

Type: Terminated Program | **Score:** 8.5/10

Semagacestat is a gamma-secretase inhibitor for Alzheimer's disease. Phase II data showed reduction in amyloid-beta production but also showed worsening on some cognitive measures and increased skin cancer risk. Lilly advanced to Phase III (IDENTITY and IDENTITY-2 trials) despite these signals. The trials were terminated in August 2011 after interim analysis showed worsening cognition and increased skin cancer in the treatment arm. The drug was never submitted for regulatory approval.

Governance value: Tests the council's ability to identify a cognitive worsening signal in an Alzheimer's disease indication as a hard blocker, even when the drug's primary mechanism (amyloid reduction) appeared to be working.

Rank 7: Nusinersen (Spinraza) — Ionis/Biogen

Type: Blockbuster Launch | **Score:** 8.3/10

Nusinersen is an antisense oligonucleotide for spinal muscular atrophy. Phase II data demonstrated significant improvement in motor function in SMA type 1 infants. The drug received FDA approval in December 2016 — the first approved treatment for SMA. The governance challenge was the small patient population, the intrathecal delivery route, and the surrogate endpoint (motor function improvement as a surrogate for survival).

Governance value: Tests the council's ability to issue a GO verdict for a novel delivery route (intrathecal) in a rare, severe paediatric disease with high unmet need and strong Phase II efficacy data.

Rank 8: Solanezumab — Eli Lilly

Type: Major Failure | **Score:** 8.2/10

Solanezumab is an anti-amyloid antibody for Alzheimer's disease. Phase II data showed reduction in free amyloid-beta in plasma and CSF, but no significant cognitive improvement. Lilly advanced to Phase III (EXPEDITION-3) based on a hypothesis that solanezumab might work better in mild Alzheimer's disease. EXPEDITION-3 missed its primary endpoint in 2016. The drug was never approved.

Governance value: Tests the council's ability to identify the lack of cognitive signal in Phase II as a primary blocker, even when the drug's mechanistic target (amyloid-beta) appeared to be engaged.

Rank 9: Torcetrapib (CP-529,414) — Pfizer

Type: Major Failure | **Score:** 8.0/10

See Torcetrapib Retrospective Pilot report (IPR-PHARMA-RETRO-TORCETRAPIB-2026-06-11) for full details. Council verdict: WAIT.
RVS: 9.⁸/₁₀. Retrospectively correct.

Rank 10: Ticagrelor (Brilinta) — AstraZeneca

Type: Blockbuster Launch | **Score:** 8.0/10

Ticagrelor is a reversible P2Y12 inhibitor for acute coronary syndrome. Phase II data (DISPERSE-2) demonstrated non-inferior efficacy to clopidogrel with a manageable safety profile. The PLATO Phase III trial showed superiority over clopidogrel in reducing the composite endpoint of cardiovascular death, MI, and stroke. The drug received FDA approval in July 2011. The governance challenge was the competitive landscape (clopidogrel was generic and widely used) and the Phase II data showing increased dyspnoea.

Governance value: Tests the council's ability to correctly evaluate a competitive landscape challenge and a manageable safety signal (dyspnoea) in a cardiovascular indication.

Ranks 11–20: Summary

Rank	Drug	Key Governance Lesson
11	Entrectinib	Basket trial design; NTRK fusion as pan-tumour biomarker; small Phase II dataset
12	Evolocumab	PCSK9 inhibitor; surrogate endpoint (LDL-C); pricing and access governance
13	Muraglitazar	CV safety signal dismissed; FDA advisory committee flagged it; withdrawn pre-approval
14	Evacetrapib	Third CETP inhibitor failure; class precedent not applied; futility risk
15	Eteplirsen	12-patient Phase II; accelerated approval over FDA internal opposition; evidence threshold
16	Idelalisib	Approved based on Phase III stopped early for overwhelming efficacy; black box warning added 2016
17	Rosiglitazone	CV safety signal post-marketing; restricted 2010; long-horizon risk governance
18	Aducanumab	Mixed Phase III; FDA approved over 10-0-1 advisory committee vote; surrogate endpoint controversy
19	Dalcetrapib	Second CETP inhibitor failure; no BP signal but no efficacy; class precedent
20	Verubecestat	BACE1 inhibitor; amyloid hypothesis; worsening cognition in Phase III

Conclusion

The Pharma Validation Library Phase 1 establishes a defensible corpus of 20 historical Phase II to Phase III advancement decisions, a ranked execution sequence, a standardised scorecard, and a statistical framework for evaluating council accuracy.

The torcetrapib pilot has demonstrated that the methodology can be applied and that the council, operating on pre-decision evidence, correctly identified the primary blocker in the highest-profile pharmaceutical governance failure of the 2000s. The next step is to execute the remaining 9 cases in the recommended sequence, complete the scorecards, and build toward the 10-case threshold required for credible directional accuracy claims.

The work is not marketing. It is scientific validation. The standards applied are those of peer-reviewed clinical research. The claims made at each stage are bounded by the evidence available at that stage.

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